

A Little Bit of History

It all started in the 1850's when Riemann questioned the meaning of

$$\int_a^b f(x) dx.$$

This concept of integration goes back to the classic antiquity to calculate certain areas, volumes, centers of mass, etc., i.e., the theory of integration.

The modern theory of integration took the form by which it is known today since the 17th century, with the works of Newton and Leibniz, and before them Fermat and Barrow.

This determines the birth of infinitesimal calculus: the dichotomy between integration and differentiation, basically as "inverse operations of each other", and we call the Fundamental Thm. of Calculus:

$$\bullet \quad F(x) = \int_a^x f(t) dt \Rightarrow F'(x) = f(x)$$

$$\bullet \quad F'(x) = f(x) \Rightarrow \int_a^b f(t) dt = F(b) - F(a).$$

In the beginning of the 19th century, Cauchy initiated the study of rigorous ideas central to Calculus, such as:

limit

derivative

integral

continuity, etc.

and introduced the first satisfactory notion of integral, albeit restricted to continuous functions.

In the middle of the 19th century, and motivated by questions posed by Fourier, Riemann developed the Theory of Integration:

- which functions can be represented by a Fourier series?
- how can one solve partial differential equations issuing from mathematical physics?
- what do we mean by "function"?
- and he noted that the notion of integration should be extended to some non-continuous functions...

This motivated the notion of Lebesgue integral: just as the real numbers "complement" the rational numbers, we'll see that functions Lebesgue integrable "complement" those functions integrable in the sense of Riemann.

Today, the Theory of (Lebesgue) Integration central to Mathematics, is intrinsically relevant to

Functional Analysis

Calculus of Variations

Differential Equations

Probability Theory,

Etc.

and is ubiquitous in Engineering, Economics, Biology, etc.

According to Lebesgue, the "geometric definition of integral" is:
the integral of a function f = measure of the
non-negative region under the graph

Now, this leads to the question of what is a measure, which sets can be attributed a measure (i.e., are measurable sets).

Consider the problem of measurement of length of subsets of $\mathbb{R}$, e.g., we aim at defining the length μ of subsets of $I := [0, 1]$.

If $J \subset I$ is an interval of the form (a, b) , $[a, b)$, $(a, b]$ or $[a, b]$,

we set $\mu(J) := |b - a|$.

For a finite union of disjoint intervals $J_1, \dots, J_n$, set

$$\mu\left(\bigcup_{i=1}^n J_i\right) = \sum_{i=1}^n \mu(J_i).$$

A

This collection of sets is called the collection of elementary sets. (later on, we will restrict the notion of elementary sets to intervals)

How can we extend the measure to non-elementary sets?

Going back to antiquity, a natural idea is to approximate non-elementary sets by elementary sets.

But using what notion of approximation?

The construction that leads to the so called Jordan measure (or Peano-Jordan measure) is this:

a set $A \subset I$ is Jordan measurable if $\forall \varepsilon > 0$
 $\exists$ elementary sets $A_\varepsilon, B_\varepsilon$ s.t.

$$A_\varepsilon \subset A \subset B_\varepsilon \text{ and } \mu(B_\varepsilon \setminus A_\varepsilon) < \varepsilon$$

(N.B. $A_\varepsilon, B_\varepsilon$ elementary sets $\Rightarrow B_\varepsilon \setminus A_\varepsilon$ elementary set).
 $\mu(B_\varepsilon \setminus A_\varepsilon) = \mu(B_\varepsilon) - \mu(A_\varepsilon)$.

When $\varepsilon \rightarrow 0$, $\mu(B_\varepsilon)$ and $\mu(A_\varepsilon)$ will have a common limit, which we'll define as $\mu(A)$.

With this procedure many sets are not Jordan measurable. Indeed, $\mathbb{Q} \cap I$ contains no elementary set of positive measure, but any elementary set containing $\mathbb{Q} \cap I$ has μ -measure 1.

Still, μ has nice properties such as

• additivity: $\mu(A \cup B) = \mu(A) + \mu(B)$, if $A \cap B = \emptyset$
[can be extended to countably many] and A, B are Jordan measurable

• translation invariance: $\mu(a+A) = \mu(A)$

So a more inclusive extension of μ was proposed by Lebesgue, changing the way of approximating by elementary sets:

- introduce the outer measure $\mu^*(A)$ for every set $A \subset I$:

$$\mu^*(A) := \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : A_n \text{ elementary sets, } A \subseteq \bigcup_{n=1}^{\infty} A_n \right\}$$

We'll see (Prop 2.9) $A \text{ elem. set} \Rightarrow \mu^*(A) = \mu(A)$

Def 0.1 A set A is said to be Lebesgue measurable if

$$\mu^*(A) + \mu^*(I \setminus A) = \mu(I) \quad (\text{cf. Def 3.1 and Rmk 3.13}^*)$$

Rmk 0.2 Elementary sets are Lebesgue measurable. See Hmw # 1, (a), (*)

Equivalently, A is Lebesgue measurable iff $\forall \varepsilon > 0$

$\exists A_\varepsilon$ elementary set st.

$$\mu^*(A \Delta A_\varepsilon) < \varepsilon,$$

where $A \Delta B := (A \setminus B) \cup (B \setminus A)$. (symmetric difference)

N.B. unlike the Jordan measure, no inclusion of elementary sets is required.

This "wance" leads to a substantial enlargement of the class of measurable sets, to the extent that the existence of non-Lebesgue measurable sets is anchored in deep set-theoretic axioms (e.g., axiom of choice).

It is then a fact that countable unions and intersections of Lebesgue measurable sets are still Lebesgue measurable, complements of measurable sets are measurable, countable additivity still holds, etc.

N.B. Countable additivity is a necessity. Think about approximating a disc by unions of rectangles...

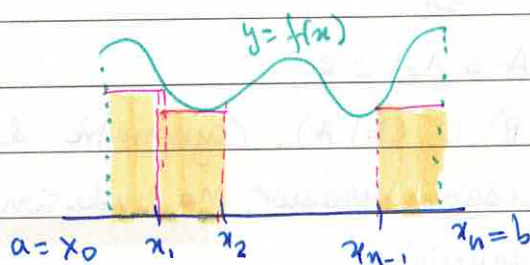
6 Lebesgue Measure of a set: in $\mathbb{R}$ ~ length; in $\mathbb{R}^2$ ~ area; in $\mathbb{R}^3$ ~ volume.

1. A Brief Overview of the Riemann Integral

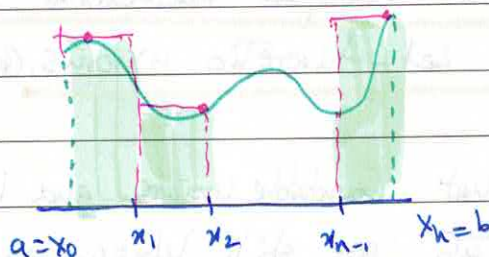
Def 1.1 Let $a, b \in \mathbb{R}$, $a < b$. A partition of $[a, b]$ is a set of $x_0, \dots, x_n$, $n \in \mathbb{N}$, with $x_i \neq x_j$ if $i \neq j$, and $x_0 = a < x_1 < \dots < x_n = b$.

Def 1.2 Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function, and let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$.

The lower Riemann sum is

$$L(f, P, [a, b]) := \sum_{i=1}^n (x_i - x_{i-1}) \inf_{[x_{i-1}, x_i]} f$$


The upper Riemann sum is

$$U(f, P, [a, b]) := \sum_{i=1}^n (x_i - x_{i-1}) \sup_{[x_{i-1}, x_i]} f$$


N.B. 1.3 Clearly $L(f, P, [a, b]) \leq U(f, P, [a, b])$

Def 1.4 let P and P' be partitions of $[a, b]$. P' is said to be finer than P if $P \subseteq P'$ (P' has more points).

Prop 1.5 let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function, and let P and P' be partitions of $[a, b]$, with P' finer than P . Then

$$L(f, P, [a, b]) \leq L(f, P', [a, b]) \leq U(f, P', [a, b]) \leq U(f, P, [a, b])$$

In particular, if P and P' are arbitrary partitions of $[a, b]$, then

$$L(f, P, [a, b]) \leq U(f, P', [a, b]) \quad (*)$$

Proof let $P = \{x_0, \dots, x_n\}$ and $P' = \{x'_0, \dots, x'_N\}$, $N > n$.

$\forall j = 1, \dots, n$, let $k \in \{0, \dots, N-1\}$ st.

$$x_{j-1} = x'_k < x'_{k+1} < \dots < x'_{k+m} = x_j$$

for some $m \geq 1$ st. $k+m \leq N$.

$$\text{Then } (x_j - x_{j-1}) \inf_{[x_{j-1}, x_j]} f = \sum_{i=1}^m (x'_{k+i} - x'_{k+i-1}) \inf_{[x'_{k+i-1}, x'_{k+i}]} f$$

(telescoping sum)

$$\leq \sum_{i=1}^m (x'_{k+i} - x'_{k+i-1}) \inf_{[x'_{k+i-1}, x'_{k+i}]} f$$

and thus $L(f, P, [a, b]) \leq L(f, P', [a, b])$.

Similarly, $U(f, P', [a, b]) \leq U(f, P, [a, b])$, and use 1.3.

let now P and P' be arbitrary partitions of $[a, b]$ and let $P'' := "P \cup P'",$ the partition that contains all points of P and all points of P' .

Then P'' is finer than P and P' , so

$$L(f, P, [a, b]) \leq L(f, P'', [a, b]) \leq U(f, P'', [a, b]) \leq U(f, P, [a, b]).$$

Def 1.6 let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function.

The lower Riemann integral of f is

$$L(f, [a, b]) := \sup_{P \text{ partition}} L(f, P, [a, b]).$$

The upper Riemann integral of f is

$$U(f, [a, b]) := \inf_{P \text{ partition}} U(f, P, [a, b]).$$

Remark 1.7 By Prop 1.5 (*)

$$L(f, [a, b]) \leq U(f, [a, b]).$$

Def 1.8 let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function.

f is said to be Riemann integrable if

$$L(f, [a, b]) = U(f, [a, b]),$$

in which case we define

$$\int_a^b f := L(f, [a, b]) = U(f, [a, b]).$$

(or $\int_a^b f(x) dx$)

Remark 1.9 $(b-a) \inf_{[a, b]} f \leq \int_a^b f \leq (b-a) \sup_{[a, b]} f.$

It can be easily proved that

Proposition 1.10 let $f, g: [a, b] \rightarrow \mathbb{R}$ be bounded Riemann integrable functions. Then $\alpha f + \beta g$ is Riemann integrable for all $\alpha, \beta \in \mathbb{R}$ and

(linearity)

$$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g.$$

Thm 1.11 If $f: [a, b] \rightarrow \mathbb{R}$ is continuous then f is Riemann integrable.

Proof Since f is continuous on a closed, bounded interval, then f is bounded and uniformly continuous.

let $\varepsilon > 0$ and choose $\delta > 0$ such that

$$s, t \in [a, b], |s - t| < \delta \Rightarrow |f(s) - f(t)| < \varepsilon. \quad (*)$$

let $n \in \mathbb{N}$ be such that $\frac{b-a}{n} < \delta$, and choose an equally spaced partition $P = \{x_0 = a, \dots, x_n = b\}$ with

$$x_j - x_{j-1} = \frac{b-a}{n}, \quad j = 1, \dots, n.$$

Then

$$\begin{aligned} U(f, [a, b]) - L(f, [a, b]) &\leq U(f, P, [a, b]) - L(f, P, [a, b]) \\ &= \frac{b-a}{n} \sum_{i=1}^n \left(\sup_{[x_{j-1}, x_j]} f - \inf_{[x_{j-1}, x_j]} f \right) \\ &\stackrel{(*)}{\leq} \frac{b-a}{n} n \varepsilon = (b-a) \cdot \varepsilon. \end{aligned}$$

letting $\varepsilon \rightarrow 0^+$ we conclude that $U(f, [a, b]) \leq L(f, [a, b])$, hence $\square$.

Why the Riemann integral is not good enough...

- Riemann integration does not handle functions with many discontinuities
- Riemann integration does not handle unbounded functions
- Riemann integration does not work well with limits:
 $f_n \rightarrow f$ (in some sense) $\Rightarrow \int_a^b f_n \Rightarrow \int_a^b f$

Example 1.12 [Discontinuous Function]

$$f: [0,1] \rightarrow \mathbb{R}, \quad f(x) := \begin{cases} 1 & x \in \mathbb{Q}, \\ 0 & x \in [0,1] \setminus \mathbb{Q}. \end{cases}$$

If $[a,b] \subset [0,1]$, $a < b$, then $\inf_{[a,b]} f = 0$, $\sup_{[a,b]} f = 1$

Thus $L(f, P, [0,1]) = 0$, $U(f, P, [0,1]) = 1 \quad \forall P \text{ partition}$,

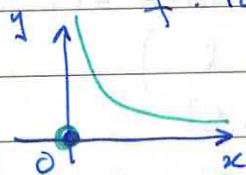
hence $L(f, [0,1]) = 0 < U(f, [0,1]) = 1$.

We conclude that f is not Riemann integrable.

(see Example 9.3* ... it is Lebesgue integrable)

Example 1.13 [Unbounded functions]

$$f: [0,1] \rightarrow \mathbb{R}, \quad f(x) := \begin{cases} \frac{1}{\sqrt{x}} & \text{if } 0 < x \leq 1, \\ 0 & \text{if } x = 0. \end{cases}$$

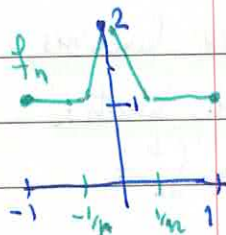


Let P be a partition of $[0,1]$. Then $\sup_{[x_{i-1}, x_i]} f = +\infty$

$$\Rightarrow U(f, P, [0,1]) = +\infty.$$

However, we should consider the area under the graph of f to be 2 and not ∞ , because

$$\lim_{a \downarrow 0} \int_a^1 f = \lim_{a \downarrow 0} \left[2\sqrt{x} \right]_a^1 = \lim_{a \downarrow 0} (2 - 2\sqrt{a}) = 2.$$



$f_n \rightarrow f$ point.
not unif.

$$f(x) = \begin{cases} 1 & x \neq 0 \\ 2 & x = 0 \end{cases}$$

Def 1.14 Let X be a set, and let $f_n, f: X \rightarrow \mathbb{R}$.

$f_n \rightarrow f$ pointwise if $\forall x \in X \quad \lim_{n \rightarrow \infty} f_n(x) = f(x)$, i.e.

$$\forall x \in X \quad \forall \varepsilon > 0 \quad \exists n = n(\varepsilon, x) \quad \forall n \geq n_0 \quad |f_n(x) - f(x)| < \varepsilon.$$

$f_n \rightarrow f$ uniformly if $\forall \varepsilon > 0 \quad \exists n_0 = n_0(\varepsilon) \quad \forall n \geq n_0 \quad \forall x \in X$

$$|f_n(x) - f(x)| < \varepsilon.$$

Recall: the uniform limit of continuous functions is also continuous